

Krishiv Patel

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Education

Rutgers University, School of Engineering

B.S. Aerospace Engineering, Minor in Mathematics, Energy Systems Certificate

Piscataway, NJ
Expected May 2027

- GPA 3.44/4.00 | Dean's List: 4 semesters
- James J. Slade Scholar: graduate research program, 6 credits of graduate independent research (16:650:542, 16:650:543)
- Honors: Francisco Diaz-Masso Engineering Scholarship, NASA Space Grant Scholarship, Rutgers School of Engineering Scholarship
- NASA L'SPACE Mission Concept Academy, Fall 2026: thermal subsystem engineer taking a spacecraft mission concept through a NASA-reviewed PDR

Experience

NASA New Jersey Space Grant Consortium – Rutgers University

Experimental Aerodynamics Research Intern

Piscataway, NJ
May 2026 – Aug 2026

- Directed five Governor's School of Engineering & Technology scholars, from the top 64 high school engineers in New Jersey, on a NASA-funded study open only to James J. Slade Scholars
- Built the test setup, calibration procedure, and data reduction workflow; ran CAD, CFD, 3D printing, and wind tunnel testing across 51 biplane configurations at $Re = 20,100$
- Peak L/D 9.12 across four trials vs 4.21 monoplane; corresponding author on both resulting papers

Rutgers Department of Mechanical & Aerospace Engineering

Research Lead, Design & CFD (from Undergraduate Researcher)

Piscataway, NJ
Mar 2025 – Present

- Lead design on a \$42K pulsed-air actuator replacing fixed nacelle chines with deployable active flow control, 150–200 m/s jets at 10–1000 Hz
- Re-architected after a manufacturability review with Calspan and GE engineers: a localized plenum replaced the external compressor feed, decoupling pressure from modulation; aerodynamics reviewed by two former Boeing engineers including a Technical Fellow
- Own CFD (ANSYS Fluent, CFX) and parametric CAD (SolidWorks); tolerance to ASME Y14.5-2018, informed by hands-on fabrication certification

Rutgers Rocket Propulsion Lab

Design & Aerodynamics, 750 lbf Lab Engine Program

Piscataway, NJ
Mar 2025 – Present

- Contribute to regenerative cooling design on the lab's pressure-fed N₂O/IPA engine; perform coolant channel sizing and flow analysis, with design changes yielding a 3% reduction in chamber wall temperature
- Size injector plate orifices to maintain thermal separation between regenerative coolant passages and the solid rocket motor igniter, translating hand calculations into geometry revisions
- Support the program through its hot fire demonstration in May 2026

Rutgers Department of Mechanical & Aerospace Engineering

Wind Tunnel Lab Developer

Piscataway, NJ
Aug 2026 – Present

- Co-developing the wind tunnel experiment being written into the curriculum of the Rutgers MAE Senior Lab, a required course for all Aerospace and Mechanical seniors
- Authored the test methodology, calibration procedure, data reduction workflow, and documented limitations

Projects

750 lbf IPA/N₂O Bipropellant Rocket Engine (*Head of Design, Rutgers Senior Design*)

May 2026 – Present

- Lead design across an 8-person team and own injector plate design: converted element geometry from coaxial swirl to shear coaxial for machinability and uniform propellant distribution across the injector face
- Diagnosed prior-generation engine: 50 lbf measured against a 750 lbf target, traced to unsustained chamber ignition driving combustion downstream of the throat with no choked flow; redesigning igniter integration to establish sustained in-chamber combustion
- Own vehicle integration and the static test stand for cold-flow characterization and hot fire, including feed system design and high-pressure procedures

Hypersonic Nose Cone Aerothermal Analysis (*University Consortium for Applied Hypersonics*)

Jul 2025 – Dec 2025

- Analyzed multiple nose cone geometries at hypersonic conditions to characterize shockwave structure and aerodynamic heating, splitting flow solution in ANSYS Fluent with conduction analysis in ANSYS Mechanical (Transient Thermal)
- Conducted blunt vs sharp trade study balancing drag against stagnation heating; integrated an internal honeycomb pattern for conductive heat dissipation

Publications

Experimental Investigation of Airfoil Aerodynamics and Flow Control Concepts: Mono and Biplane Configurations

- IEEE MIT Undergraduate Research Technology Conference, accepted; presenting Oct 2026 | co-corresponding author with Dr. DeMauro
- GSET Research Journal, published July 2026 | sole corresponding author

Technical Skills

Analysis: ANSYS Fluent, ANSYS CFX, ANSYS Mechanical (Transient Thermal), XFLR5, MATLAB, Python, LabVIEW; classical lamination theory, composite failure criteria (Hashin, max stress)

Design: SolidWorks, OnShape, ANSYS SpaceClaim, GD&T (ASME Y14.5-2018), Design for Manufacturability

Experimental & Fabrication: Wind tunnel testing and calibration, hot-wire anemometry, smoke flow visualization, Shopbot CNC, laser cutting, FDM 3D printing